AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MS102
Physics

Final Examination
Summer Semester 2025/2026

Time Allowed: 2hrs

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Instructions:

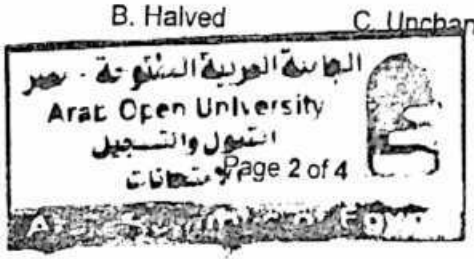
- Total Marks: **50**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART I: Answer the entire Multiple Choice Questions (MCQ). [25 Marks]

1. A particle moves with constant acceleration. The distances covered during two successive 2 s intervals are 12 m and 28 m. The acceleration is
   A. 2 m/s² B. 3 m/s² C. 4 m/s² D. 5 m/s²
2. A string of linear density 4.0×10^{-3} kg/m is under a tension of 100 N. The speed of waves on the string is closest to
   A. 79 m/s B. 112 m/s C. 158 m/s D. 200 m/s
3. Two charges of +6 μC and −2 μC are separated by 0.30 m. If both charges are doubled and the separation is tripled, the force becomes
   A. 4F/9 B. 2F/3 C. 8F/9 D. 4F/3
4. In the circuit defined by Kirchhoff's loop rule, the algebraic sum of all potential differences around a closed loop must equal
   A. The total current B. Zero
   C. The equivalent resistance D. The emf only
5. A 0.20 kg ball moving at 15 m/s strikes a wall and rebounds with speed 10 m/s. The magnitude of the impulse delivered to the ball is
   A. 1 N·s B. 2 N·s C. 5 N·s D. 10 N·s
6. A standing wave is produced on a string fixed at both ends. If the string length equals $3\lambda/2$, the harmonic number is
   A. 2 B. 3 C. 4 D. 5
7. A body projected vertically upward with speed 39.2 m/s reaches a maximum height of approximately
   A. 39.2 m B. 58.8 m C. 78.4 m D. 98.0 m
8. Three resistors 2 Ω, 3 Ω, and 6 Ω are connected in parallel. The equivalent resistance is
   A. 0.5 Ω B. 1.0 Ω C. 1.5 Ω D. 2.0 Ω
9. A 2 kg object initially at rest experiences a net force F=12 N. Speed after 3s is
   A. 6 m/s B. 9 m/s C. 12 m/s D. 18 m/s
10. Three identical charges +q occupy three corners of a square. The net force on a charge placed at the fourth corner is directed
    A. Along one side B. Along the diagonal away from the center
    C. Toward the center D. Perpendicular to the square
11. If the tension in a string is increased by 40%, the wave speed increases by
    A. 20% B. 30% C. 44% D. 50%
12. Two identical conducting spheres carry charges +8 μC and −2 μC. After contact and separation, the charge on each sphere is
    A. +1 μC B. +2 μC C. +3 μC D. +4 μC
13. A battery of emf 18 V and internal resistance 1 Ω supplies current to an external resistor of 5 Ω. The current is
    A. 2 A B. 3 A C. 4 A D. 5 A
14. A 4 kg cart moving at 5 m/s collides elastically head-on with an identical cart initially at rest. The speed of the second cart after collision is
    A. 0 m/s B. 2.5 m/s C. 5 m/s D. 10 m/s
15. If the frequency of a wave doubles while its speed remains constant, the wavelength becomes
    A. Doubled B. Halved C. Unchanged D. Quadrupled



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16. A charge q placed in an electric field E experiences a force F. If both q and E are doubled, the force becomes
A. F B. 2F C. 4F D. 8F

17. In a circuit, a 12 Ω resistor dissipates 48 W. The current through it is
A. 1 A B. 2 A C. 3 A D. 4 A

18. A particle moving at 10 m/s accelerates uniformly at 4 m/s². The distance covered in the next 3 s is
A. 36 m B. 42 m C. 48 m D. 54 m

19. Which of the following waves is necessarily longitudinal?
A. Electromagnetic waves B. Waves on a stretched string
C. Sound waves in air D. Water surface waves

20. Two charges separated by 0.50 m exert a force of 8 N. If the separation becomes 1.0 m, the force is
A. 1 N B. 2 N C. 4 N D. 8 N

21. Using Kirchhoff's junction rule, if 7 A enters a junction and currents of 2 A and 3 A leave, the remaining current leaving is
A. 1 A B. 2 A C. 3 A D. 4 A

22. A body falls freely from rest. The ratio of distances covered in the first and second seconds is
A. 1:2 B. 1:3 C. 1:4 D. 2:3

23. A bullet of mass 0.010 kg moving at 400 m/s passes completely through a block of mass 1.0 kg initially at rest. The bullet emerges at 100 m/s. The speed acquired by the block is
A. 1 m/s B. 2 m/s C. 3 m/s D. 4 m/s

24. The maximum speed of a wave on a string can be increased by
A. Increasing linear density B. Decreasing tension
C. Increasing tension D. Shortening wavelength only

25. A resistor R is connected across a fixed voltage source. If the resistance is doubled, the power dissipated becomes
A. P/4 B. P/2 C. 2P D. 4P

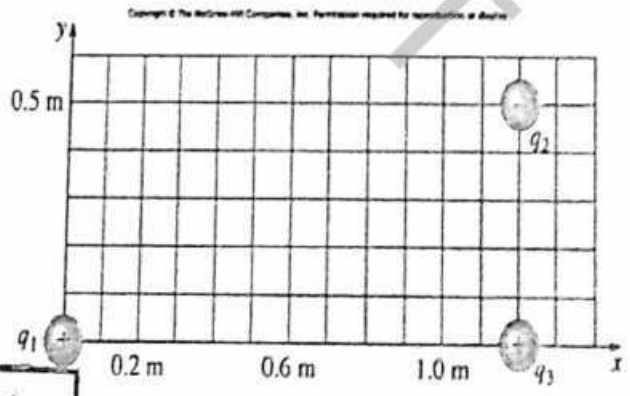
Part II: Answer the following questions. [25 Marks]

Question1 (5 Marks)

A string is stretched with a tension of 120 N. The string has a length of 2.5 meters. The velocity of wave propagation along the string is 220 m/s. Calculate the mass per unit length of the string is?

Question2 (10 marks)

What is the net force on the charge q_1 due to the other two charges?
$q_1 = +1.2\ \mu C$, $q_2 = -0.60\ \mu C$, and $q_3 = +0.20\ \mu C$.
Distance between q_1 and q_3 =1.2m
Distance between q_2 and q_3 is 0.5m

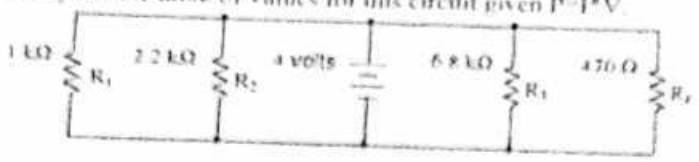


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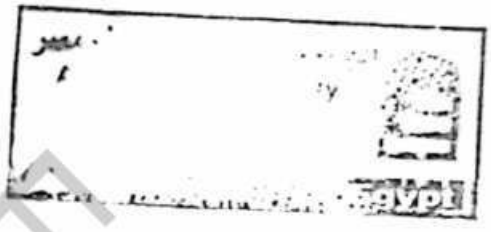
Question3 (10 Marks)

Complete the table of values for this circuit given P=I*V



	R_1	R_2	R_3	R_4	Total
V					
I					
R	1 kΩ	2.2 kΩ	6.8 kΩ	470 Ω	
P					

End of questions



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